



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SARASWATHI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

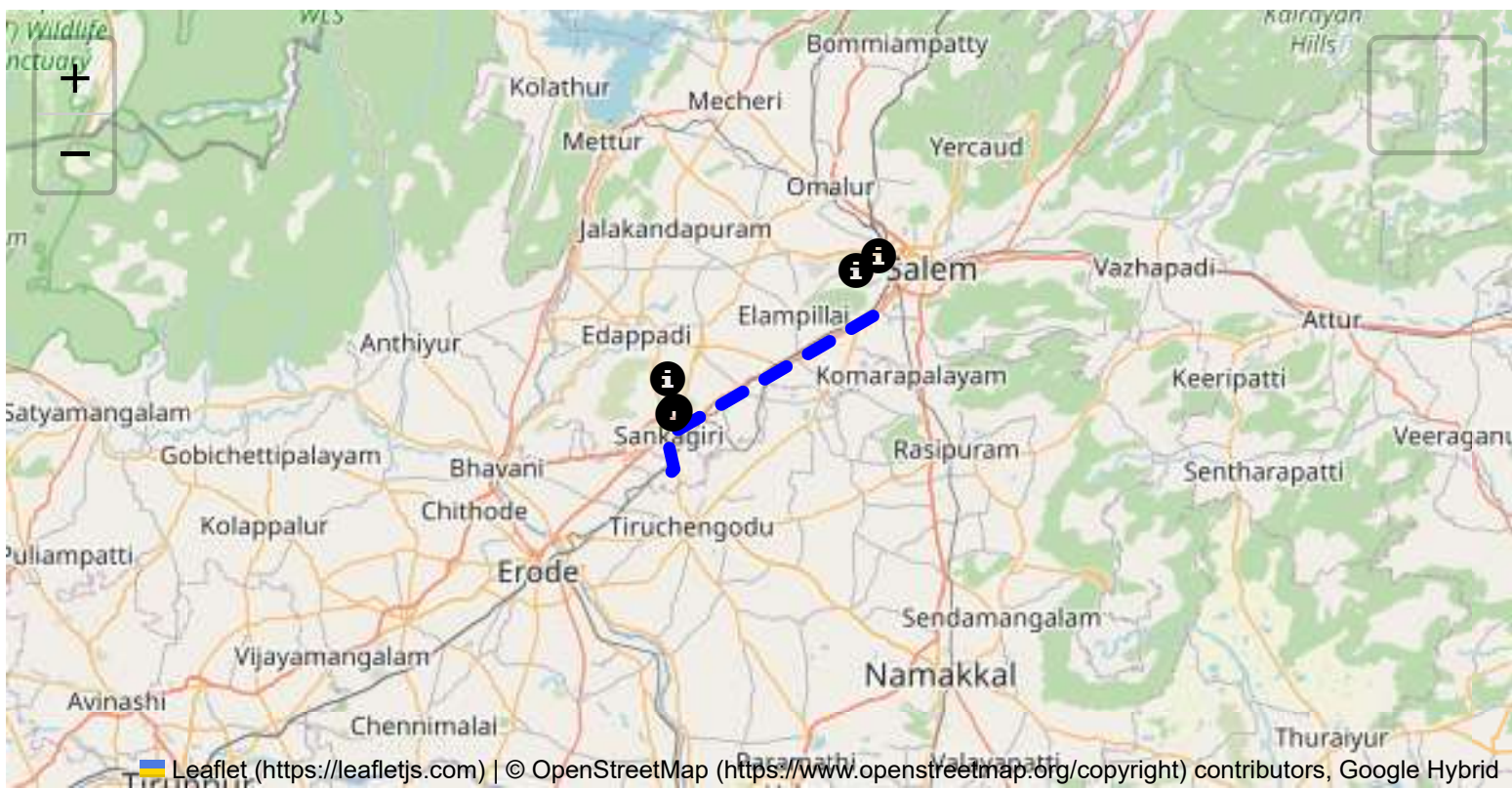
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

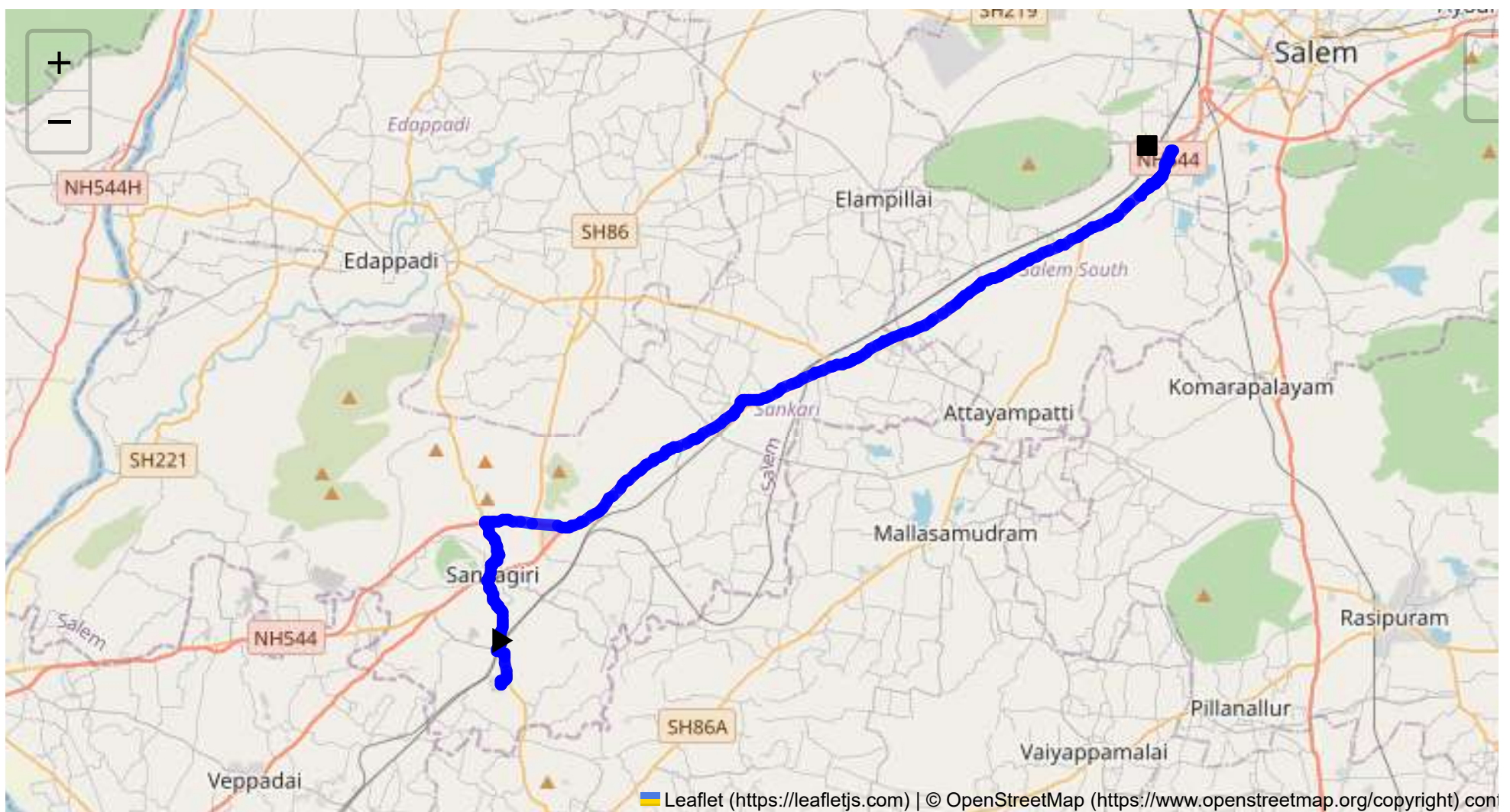
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 39.97 km
Estimated Duration: 0.8 hours
Adjusted Duration (Heavy Vehicle): 1.0 hours
Start: (11.4381, 77.8734)
End: (11.607557, 78.098218)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route starts at CVQF+23W, Sangagiri and ends at Uthamasolapuram, traversing through Kangeyampalayam and Salem. The journey spans approximately 39.97 kilometers, primarily along NH-544 and SH-86, known for facilitating regional connectivity. The route features urban, semi-urban, and rural sections, each associated with distinct infrastructure and driving conditions.

2. Weather Conditions

Tamil Nadu typically experiences tropical weather. The weather along this route varies:

- **Summer (March to June):** High temperatures, occasionally exceeding 40°C, which might lead to road surface issues like softening of asphalt.
- **Monsoon (July to September):** Heavy rainfall, causing potential waterlogging and reduced visibility. Roads could become slippery.
- **Winter (November to February):** Mild and pleasant, posing fewer weather-related hazards.

3. Traffic Patterns

- **Peak Hours:** Morning (8 AM - 10 AM) and evening (5 PM - 8 PM) experience higher traffic, especially near Salem due to commuter traffic.
- **Congestion-Prone Areas:** Market areas in Salem and near major junctions such as the NH-544 interchange in Sangagiri are bottlenecks.

4. Road Quality and Infrastructure

- **Road Conditions:** Generally good, but some patches need maintenance, especially along rural stretches.
- **Infrastructure Challenges:** Narrow stretches and sharp curves in rural zones may require extra caution when steering large vehicles.

5. Alternative Routes

In the event of emergencies or roadblocks, consider detouring via SH-86 to Bhavani and rejoining at Salem. This route avoids the NH-544 segment between Sangagiri and Kangeyampalayam, providing an alternative while adding minimal extra distance.

6. Local Regulations for Hazardous Materials

Carrying hazardous materials demands adherence to:

- Specific time restrictions, typically avoiding peak hours.
- Mandates to display HAZMAT signage.
- Compliance with speed limits and restricted parking zones.

7. Historical Incidents

No major incidents involving hazardous materials have been recently reported. However, minor accidents occasionally occur on the congested segments of NH-544, predominantly due to human error or vehicular failure.

8. Environmental Considerations

- **Sensitive Areas:** Passes near agricultural zones; ensure no spillage or contamination during transit.
- **Protected Regions:** None directly along the route, but conservation efforts in adjacent areas necessitate cautious transport practices.

9. Communication Coverage

Mobile network and communication coverage are robust along most parts of the route, with potential dead zones more likely in rural regions. Kangeyampalayam has occasional network issues.

10. Emergency Response Times

- **Urban segments (Salem, Sangagiri):** 10-15 minutes due to proximity to emergency services.
- **Rural segments (Kangeyampalayam):** 30 minutes or more, due to distance from main facilities.

12. Overall Summary of Risk Assessment

The route is generally safe for transporting heavy vehicles with hazardous materials, provided drivers remain diligent about:

- Adjusting to weather conditions.
- Navigating traffic-congested areas during off-peak times.
- Keeping aware of infrastructure limitations.

Adherence to local hazardous material transportation regulations is crucial, and drivers should be equipped with the latest communication technology to handle potential dead zones. Familiarizing oneself with emergency procedures and alternative routes enhances preparedness and minimizes risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44028, 77.87541	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	High	11.45007, 77.87201	15 KM/Hr
6	Turn	High	11.47407, 77.86839	15 KM/Hr
7	Turn	High	11.47451, 77.86914	15 KM/Hr
8	Turn	Medium	11.47920, 77.86989	30 KM/Hr
9	Turn	Medium	11.48063, 77.87165	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.48258, 77.87250	30 KM/Hr
11	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
12	Turn	Medium	11.49431, 77.87032	30 KM/Hr
13	Turn	Medium	11.49420, 77.87051	30 KM/Hr
14	Turn	Medium	11.60942, 78.10056	30 KM/Hr
15	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
16	Turn	High	11.62130, 78.10776	15 KM/Hr
17	Turn	Medium	11.61801, 78.10639	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
2	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
3	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
4	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
7	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47451, 77.86914



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47920, 77.86989



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48063, 77.87165



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48258, 77.87250



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49390, 77.86748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49431, 77.87032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49420, 77.87051



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60942, 78.10056



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61801, 78.10639
